Temporal Analysis in Google Earth Engine

Study Area: Bushfire in Northern Australia 2023

Code Source: https://gist.github.com/rabinatwayana/9274dbb1ec375df5c0f1db6bf59b8d9b

Earth Engine App Link: https://ee-rabinatwayana123.projects.earthengine.app/view/fire-in-northern-australia-2023

1. Background

According to research conducted by Fisher (2024), catastrophic bushfires that was occured in northern Australia in 2023 which burned more than 84 million hectares of desert and savannah. This bushfire began in June 2023 and lasted for several months. Thus, this case is undertaken for this assignment.

The case study is carried out using the Sentinel 2 Level 2A imageries that allows to visualize and analyze the temporal changes in burned area impacted by fire spread. To cover the entire northern Australia region, image mosaicing was done to stitch the images seamlessly.

The date-slider is provided which facilitate to analyze the changes in bushfire over the week. In addition to this, selecting the data for different week and changes can be compared using the split screen.

Results

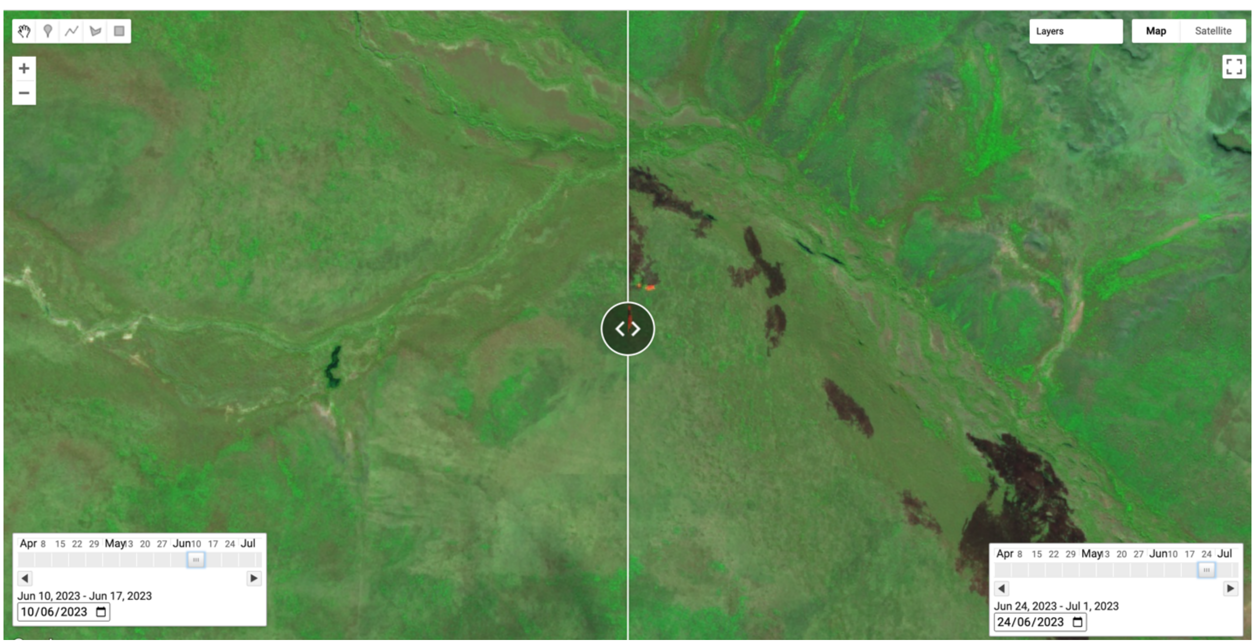

Figure 1: Split screen window with date-slider for temporal changes analysis

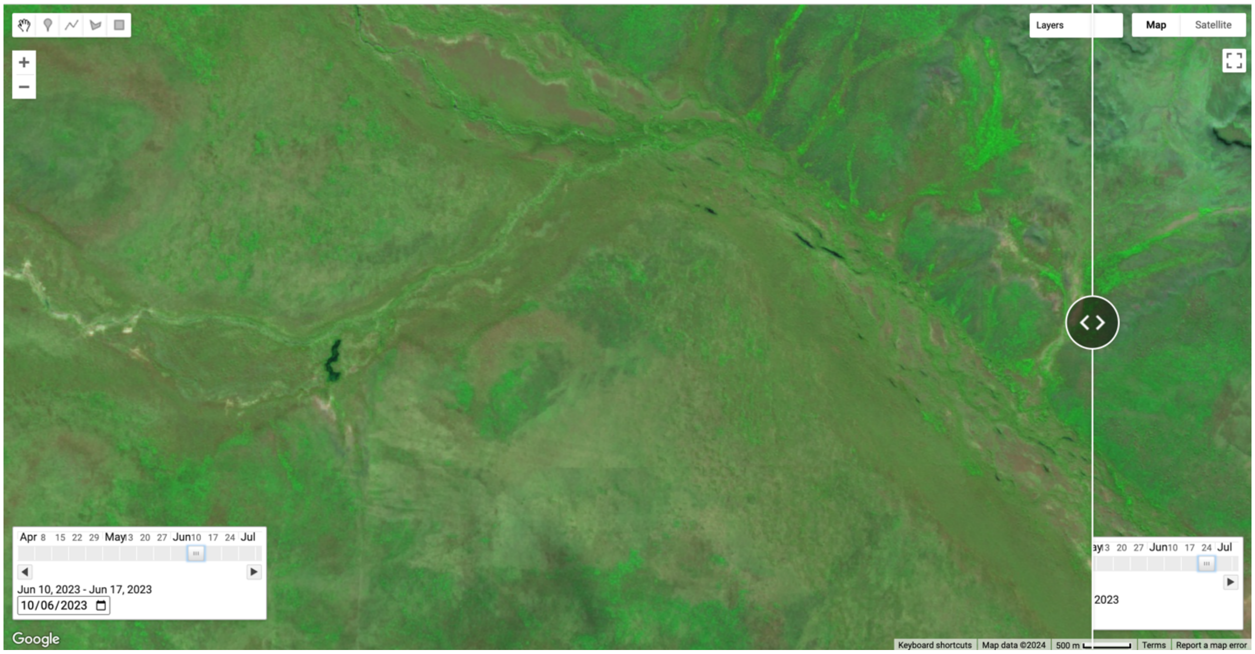

Figure 2: Area before bushfire

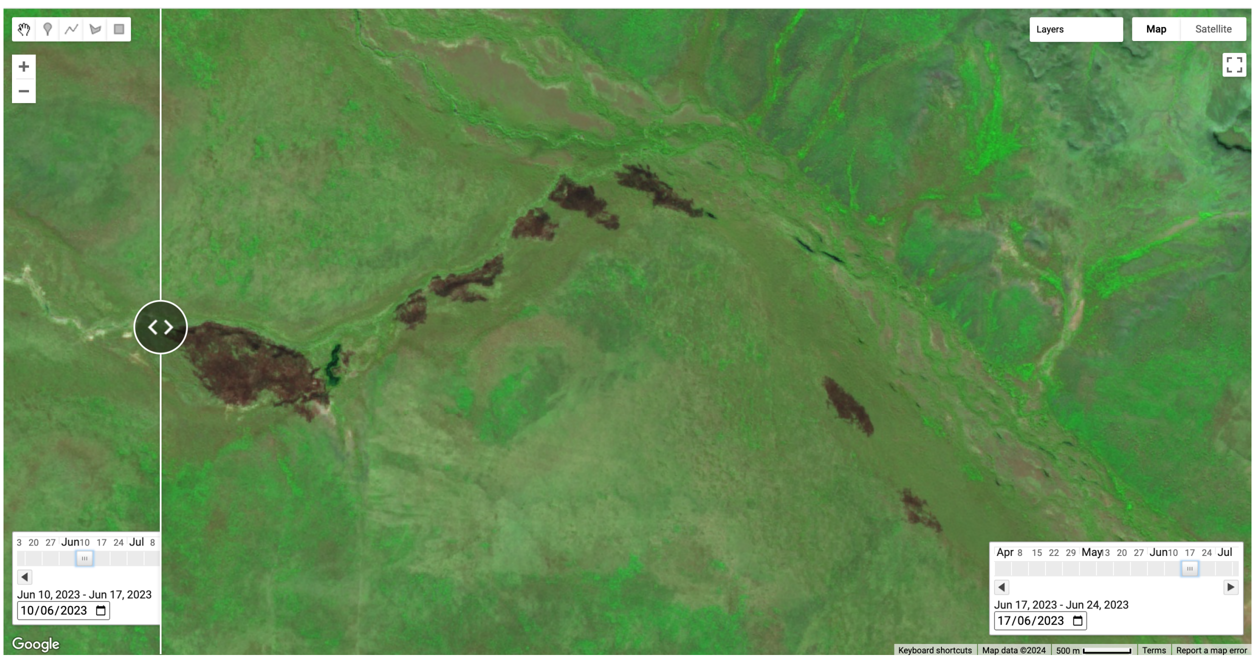

Figure 3: Area burned during Jun 17 to Jun 24, 2023

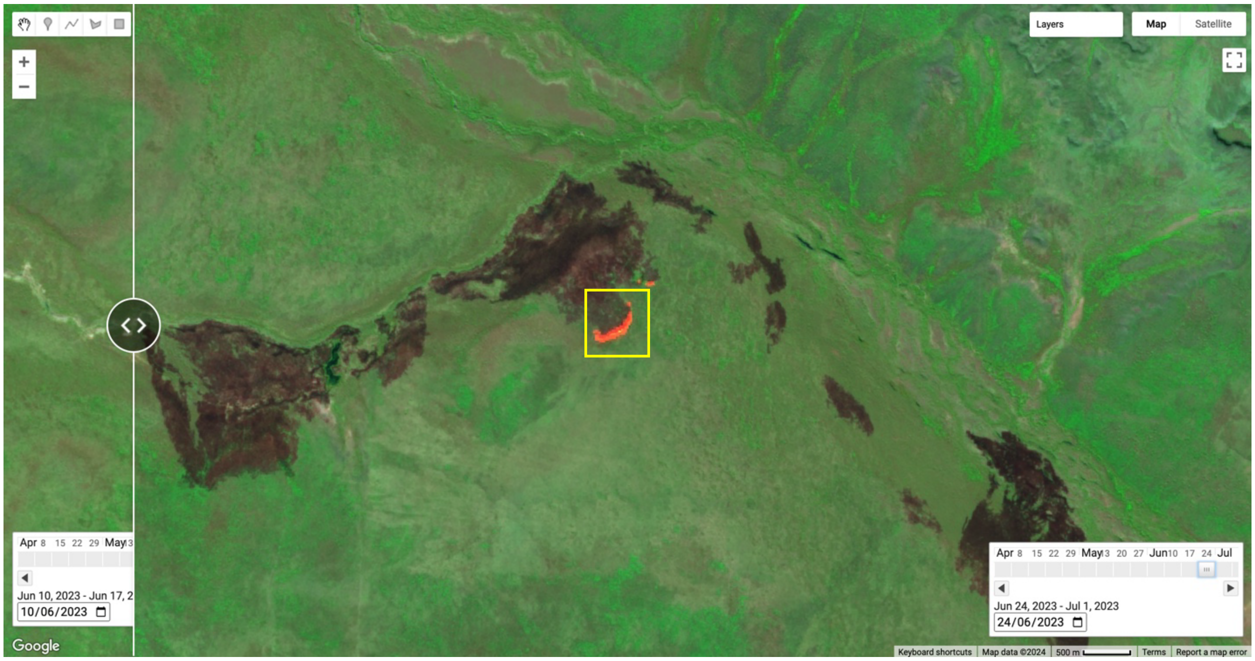

Figure 4: Fire spread and increase in burned area between Jun 24 and Jul 1 2023

Discussion

All the brown patches visible in the images (Figure 3 and Figure 4) represents burned areas whereas the active fire (indicated by yellow box), was observed in late June as represented in Figure 4. This visualization focuses on certain area of Northern Australia, but changes can be compared for other parts or region similarly.

References

Fisher, R. (2024, 04 23). *Vastly bigger than the Black Summer: 84 million hectares of northern Australia burned in 2023.* Retrieved from The Conversation Media: https://theconversation.com/vastly-bigger-than-the-black-summer-84-million-hectares-of-northern-australia-burned-in-2023-227996